

Sea-Level Rise Forecast Models Related to SPECTRUM: A Literature Assessment

Literature search compiled 2026-05-10; updated 2026-09-03; updated 2026-09-13

This report identifies published frameworks and studies most relevant to SPECTRUM, our Bayesian, observation-calibrated, component-level GMSL forecasting framework. The original version of this report (2026-05-10) covered only the six closest component-summation frameworks (Section 1). The 2026-09-03 update broadens the search to five additional methodological facets of SPECTRUM: alternative component-summation frameworks (Section 2), the semi-empirical GMST-sensitivity lineage (Section 3), deep-uncertainty and WAIS tail-risk framing (Section 4), observational anchoring and naive extrapolation (Section 5), and reduced-complexity discharge/calving parameterizations analogous to our Greenland model (Section 6). Section 7 lists items published in approximately the last twelve months. A consolidated summary and a set of specific action items for the manuscript close the report (Section 8).

1 Core component-summation frameworks (2026-05-10 search)

1.1 Mengel et al. (2016)

Citation: Mengel, M., Levermann, A., Frieler, K., Robinson, A., Marzeion, B., & Winkelmann, R. (2016). Future sea level rise constrained by observations and long-term commitment. *Proc. Natl. Acad. Sci.*, 113(10), 2597–2602.

DOI: 10.1073/pnas.1500515113

Approach. Each component (thermal expansion, glaciers, Greenland, Antarctica) is represented by a simple physically motivated response function with a small number of parameters, independently calibrated to 20th-century observations via Bayesian inference. Projections are generated by sampling posteriors and forcing with RCP temperature scenarios, then summing components. Inter-component correlations arise only through shared temperature forcing.

Relevance: Very high. This is the closest published methodological parallel: per-component semi-empirical models, each observation-calibrated, summed probabilistically. The main differences are (i) our use of rate-space and level-space Bayesian models rather than equilibrium response functions, (ii) our explicit deep-uncertainty treatment of WAIS, and (iii) our use of a blending step that anchors projections to satellite-era observed rates.

1.2 Kopp et al. (2014)

Citation: Kopp, R.E., Horton, R.M., Little, C.M., Mitrovica, J.X., Oppenheimer, M., Rasmussen, D.J., Strauss, B.H., & Tebaldi, C. (2014). Probabilistic 21st and 22nd century sea-level projections at a global network of tide-gauge sites. *Earth's Future*, 2(8), 383–406.

DOI: 10.1002/2014EF000239

Approach. The foundational probabilistic component-summation framework. Individual components (thermal expansion, glaciers, GrIS, AIS, land water storage) are each represented as prob-

ability distributions drawn from IPCC AR5 expert assessment and process models. Monte Carlo sampling sums components at each location, applying gravitational fingerprints for regional projections.

Relevance: High. Established the “plug-and-play” component-summation architecture that all subsequent probabilistic frameworks build on, including IPCC AR6. Our work shares this architecture but replaces assessment-derived component PDFs with observation-calibrated Bayesian posteriors, and does not address regional patterns.

1.3 Wong et al. (2017) — BRICK

Citation: Wong, T. E., Bakker, A. M. R., Ruckert, K., Applegate, P., Slangen, A. B. A., & Keller, K. (2017). BRICK v0.2, a simple, accessible, and transparent model framework for climate and regional sea-level projections. *Geosci. Model Dev.*, 10, 2741–2760.

DOI: 10.5194/gmd-10-2741-2017

Approach. Couples simple, physically motivated models for each component (DOECLIM for thermal expansion, Marzeion-type scaling for glaciers, SMB/dynamics models for GrIS and AIS) within a shared climate state. The full system is jointly calibrated to observations using MCMC, producing posteriors that naturally capture inter-component correlations.

Relevance: High. BRICK is the only published framework that performs joint Bayesian calibration of all components simultaneously, making it the closest analogue to our approach in terms of statistical methodology. A key distinction is that our framework calibrates each component independently and handles WAIS through a structured scenario mixture rather than a process model. See also Darnell et al. (2025, Section 4.2), a BRICK-based follow-on directly relevant to our emissions-vs-tail-risk framing.

1.4 Kopp et al. (2023) — FACTS

Citation: Kopp, R. E., Garner, G. G., Hermans, T. H. J., et al. (2023). The Framework for Assessing Changes To Sea-level (FACTS) v1.0: a platform for characterizing parametric and structural uncertainty in future global, relative, and extreme sea-level change. *Geosci. Model Dev.*, 16, 7461–7489.

DOI: 10.5194/gmd-16-7461-2023

Approach. The modular software platform underlying IPCC AR6 sea level projections. Each component is a pluggable module that can use different process models or emulators. The framework propagates both parametric uncertainty (within a model) and structural uncertainty (across alternative models for the same component).

Relevance: High. FACTS is the institutional benchmark against which any new component-level framework will be compared. Our work can be understood as a complementary approach that replaces the process-model modules in FACTS with data-driven Bayesian models, trading physical detail for tighter observational constraint and transparency about what the data alone can support.

1.5 Fox-Kemper et al. (2021) — IPCC AR6 Chapter 9

Citation: Fox-Kemper, B., Hewitt, H. T., Xiao, C., et al. (2021). Ocean, Cryosphere and Sea Level Change. In *Climate Change 2021: The Physical Science Basis* (pp. 1211–1362). Cambridge University Press.

DOI: 10.1017/9781009157896.011

Approach. The authoritative component-level assessment, summing thermal expansion (CMIP6), glaciers (GlacierMIP), Greenland (ISMIP6 + emulators), Antarctica (ISMIP6 + emulators +

structured expert judgment), and land water storage. Provides both medium-confidence and low-confidence projections across all SSP scenarios.

Relevance: High. This is the primary comparison target for any new SLR projection. Our framework produces systematically higher medians and wider uncertainty ranges than IPCC AR6 medium confidence, largely because the IPCC ice sheet component relies on ISMIP6 process models that may understate dynamic mass loss. The IPCC low-confidence projections, which incorporate expert judgment, are more consistent with our results. Törnqvist et al. (2025, Section 5) is a useful counterpoint here: it shows the 1995/96 IPCC SAR mid-range projection verified to within ~ 1 cm over the following 30 years, so any claim that “IPCC projections diverge from observations” should specify AR6 specifically rather than IPCC assessments in general.

1.6 Nauels et al. (2017) — MAGICC Sea Level Model

Citation: Nauels, A., Meinshausen, M., Mengel, M., Lorbacher, K., & Wigley, T.M.L. (2017). Synthesizing long-term sea level rise projections — the MAGICC sea level model v2.0. *Geosci. Model Dev.*, 10, 2495–2524.

DOI: 10.5194/gmd-10-2495-2017

Approach. Emulates process-based responses for seven sub-components (thermal expansion, glaciers, GrIS SMB, GrIS dynamics, WAIS, EAIS, land water storage) using reduced-complexity parameterizations calibrated to CMIP5/AR5 output. Embedded within the MAGICC climate model, ensuring a self-consistent climate trajectory across components. Extends projections to 2300.

Relevance: Moderate–high. Shares the component granularity of our framework (seven sub-components) and the goal of efficient probabilistic ensemble generation. However, MAGICC calibrates to process-model output rather than directly to observations, and does not incorporate deep uncertainty in ice sheet dynamics. Our framework’s direct observational calibration and explicit WAIS scenario structure address both of these limitations. See also Nauels et al. (2025, Section 7), a recent multi-century commitment framework from an overlapping author group.

2 Additional component-summation and probabilistic frameworks

2.1 Le Bars & Drijfhout (2025)

Citation: Le Bars, D. & Drijfhout, S. (2025). Connecting the Past and the Future to Build Regional Sea-Level Projections. *Earth’s Future*, 13, e2024EF005623.

Approach. The KNMI (Netherlands) national framework builds a GMSL/regional budget (Antarctica, Greenland, glaciers, thermosteric, land water, vertical land motion) that is deliberately anchored to start in 1995 so that it overlaps and is evaluated against the observational record before extending to 2100.

Relevance: High. Of everything found in this search, this is structurally the closest published analogue to SPECTRUM’s practice of calibrating each component against its own observational record and checking budget closure against satellite altimetry. Worth a direct comparison in the framework section.

2.2 Felikson et al. (2025)

Citation: Felikson, D., Rounce, D.R., Fasullo, J., Rodriguez, A., Adhikari, S., Buzzanga, B., Dangendorf, S., Kopp, R.E., Lammers, R.B., Reager, J.T., Brinkerhoff, D., Csatho, B., Giroto, M., Hamlington, B., Ivins, E.R., Kumar, P., Larour, E., Nerem, R.S., Nowicki, S., Schlegel, N.-J.,

Tesdal, J.-E., & Weathers, M. (2025). Progress and future directions in constraining uncertainties in sea-level projections using observations. *Nature Climate Change*, 15(10), 1039–1051.

DOI: 10.1038/s41558-025-02437-4

Approach. A 22-author community perspective (co-authored by several of the researchers behind the leading process-based and observational SLR groups: Kopp, Nerem, Hamlington, Dangendorf, Nowicki, among others) reviewing the main sources of uncertainty in multi-decadal to centennial SLR projections, and arguing explicitly for using observations to narrow them. It calls for improved emulation techniques, more systematic quantification of uncertainty *structure* in both observations and models, longer observational records for the underlying physical processes, and closer collaboration across physical and social sciences, framed explicitly against the next IPCC assessment cycle.

Relevance: Very high. This is the single closest published statement of SPECTRUM’s underlying philosophy — constrain component projections with observations rather than relying on process models alone — from a broad, senior, cross-institutional author group rather than a single competing framework. It is not itself a competing forecasting method, so it does not need to be distinguished from SPECTRUM the way the Section 1 frameworks do; instead it is best cited in the introduction or discussion as the field-level motivation SPECTRUM directly answers. See Section 9 (“near-perfect conceptual match”) for a fuller explanation.

2.3 Slangen et al. (2022)

Citation: Slangen, A.B.A., Palmer, M.D., Camargo, C.M.L., Church, J.A., Edwards, T.L., Hermans, T.H.J., Hewitt, H.T., Garner, G.G., et al. (2022). The evolution of 21st century sea-level projections from IPCC AR5 to AR6 and beyond. *Cambridge Prisms: Coastal Futures*, 1, e8.

DOI: 10.1017/cft.2022.8

Approach. Traces how successive IPCC component-summation frameworks (AR5, SROCC, AR6) changed their component treatments, uncertainty structure, and headline numbers across assessment cycles.

Relevance: Moderate–high. Useful for positioning SPECTRUM against the IPCC lineage as a whole rather than against AR6 alone, and for anticipating how the next assessment cycle (which Felikson et al. 2025 explicitly addresses) is likely to evolve.

2.4 Related items noted but not detailed

Garner et al. (2018, *Earth’s Future*, DOI 10.1029/2018EF000991) is an earlier meta-comparison of the evolution of SLR projections (1980s–2018), complementary to Slangen et al. (2022). Robel, Seroussi & Roe (2019, *PNAS*, DOI 10.1073/pnas.1904822116) shows marine ice sheet instability (MISI) mathematically skews the SLR probability distribution toward high outcomes; it is discussed at length in Section 4.3 but is equally relevant here as the theoretical justification for why a component-summation posterior for WAIS should be right-skewed rather than Gaussian. Le Cozannet, Manceau & Rohmer (2017, *Environ. Res. Lett.*) apply possibility theory rather than Bayesian Monte Carlo to bound probabilistic SLR projections — a genuinely different uncertainty formalism worth a citation if the discussion engages with alternatives to Bayesian summation.

2.5 Grandey et al. (2024) — statistical fusion of existing projections

Citation: Grandey, B.S., Dauwels, J., Koh, Z.Y., Horton, B.P., & Chew, L.Y. (2024). Fusion of probabilistic projections of sea-level rise. *Earth’s Future*, 12(12), e2024EF005295.

DOI: 10.1029/2024EF005295

Approach. Statistically fuses multiple existing probabilistic GMSL projections (e.g. IPCC AR6, expert elicitations) into a single consensus posterior, rather than generating a new projection from component physics or observations.

Relevance: Moderate–high. Methodologically adjacent to SPECTRUM’s own blending step (Section 5.2), but fuses *other studies’ outputs* rather than blending an observation-anchored posterior with a model-based one. Worth a passing citation as an alternative notion of “fusion” if the manuscript discusses its blending step’s novelty.

2.6 Perrette et al. (2013) — precursor to the tide-gauge scaling approach

Citation: Perrette, M., Landerer, F., Riva, R., Frieler, K., & Meinshausen, M. (2013). A scaling approach to project regional sea level rise and its uncertainties. *Earth Syst. Dynam.*, 4(1), 11–29.

DOI: 10.5194/esd-4-11-2013

Approach. Scales each component’s regional pattern to a global-mean trajectory derived from CMIP-class temperature projections, with uncertainty propagated through the scaling coefficients rather than independent per-component Bayesian calibration.

Relevance: Moderate. Primarily useful as the direct methodological ancestor of Perrette & Mengel (2025, Section 7), already identified as a genuinely different observation-driven alternative to component summation; citing both together shows that lineage has matured from a model-scaling approach to a purely observation-driven one.

3 Semi-empirical GMST-sensitivity models

3.1 Rahmstorf (2007) and the rate-vs-level question

Citation: Rahmstorf, S. (2007). A Semi-Empirical Approach to Projecting Future Sea-Level Rise. *Science*, 315, 368–370.

DOI: 10.1126/science.1135456 — already cited in the manuscript as `Rahmstorf2007`.

Approach. Regresses the rate of GMSLR linearly on the *level* of GMST, $dS/dt = aT$, calibrated on the 20th-century instrumental record; the founding paper of the semi-empirical class the manuscript already discusses.

Citation: Vermeer, M. & Rahmstorf, S. (2009). Global sea level linked to global temperature. *PNAS*, 106, 21527–21532.

DOI: 10.1073/pnas.0907765106 — already present in `references.bib` as `Vermeer2009` but **not currently cited in the manuscript text**.

Approach. Extends Rahmstorf (2007) by adding a rate-of-warming term, $dS/dt = aT + b dT/dt$, so that GMSLR responds to both the level and the rate of change of temperature.

Relevance: Very high. This is the direct published precedent for the open rate-vs-level question already flagged in project notes (does SLR respond to T or also dT/dt ?). It should be cited alongside Rahmstorf (2007) wherever the manuscript discusses the semi-empirical class, and is the natural anchor for any explicit discussion of that open question.

3.2 Grinsted & Christensen (2021) — transient sensitivity

Citation: Grinsted, A. & Christensen, J.H. (2021). The transient sensitivity of sea level rise. *Ocean Science*, 17, 181–186.

DOI: 10.5194/os-17-181-2021 — already in the local Zotero library, not currently cited.

Approach. Reframes the rate-vs-level question again: shows that on century timescales, GMSLR rate correlates better with *accumulated (cumulative) warming* than with instantaneous GMST, defining a “transient sea-level sensitivity” (TSLs) distinct from the millennial-scale equilibrium sensitivity. The observed TSLs ($0.40 \pm 0.05 \text{ m } ^\circ\text{C}^{-1} \text{ century}^{-1}$) exceeds CMIP-based estimates (0.27–0.39). A companion paper, Grinsted et al. (2022, *Earth’s Future*, DOI 10.1029/2022EF002696, in Zotero), extends the same diagnostic to CMIP6 GCMs directly.

Relevance: High. A third, genuinely distinct framing of the level-vs-rate question (cumulative-forcing response) that the manuscript should engage with directly if it takes a position on this question, alongside Rahmstorf (2007) and Vermeer & Rahmstorf (2009).

3.3 Critiques of the semi-empirical approach

The manuscript currently attributes IPCC AR5’s “low confidence” characterization of semi-empirical models solely to Church et al. (2013, already cited as **Church2013**), but that assessment synthesized specific empirical critiques not currently cited. The most relevant, none yet in the local Zotero library:

- Holgate, S., Jevrejeva, S., Woodworth, P., & Brewer, S. (2007). Comment on “A Semi-Empirical Approach to Projecting Future Sea-Level Rise.” *Science*, 317, 1866 (DOI 10.1126/science.1140942), with Rahmstorf’s reply in the same issue (DOI 10.1126/science.1141283). Argues no robust linear GMST-rate relationship is detectable in the data used; Rahmstorf’s reply defends split-sample predictive skill.
- Orlić, M. & Pasarić, Z. (2013). Semi-empirical versus process-based sea-level projections for the twenty-first century. *Nat. Clim. Change*, 3, 735–738 (DOI 10.1038/nclimate1877), and (2015), Some Pitfalls of the Semiempirical Method Used to Project Sea Level Rise. *J. Climate*, 28, 3779–3785 (DOI 10.1175/JCLI-D-14-00696.1). Shows semi-empirical projections are highly sensitive to which dynamical terms and detrending/response-time choices are included; a “corrected” estimate converges toward process-based values.
- Schmith, T., Johansen, S., & Thejll, P. (2012). Statistical Analysis of Global Surface Temperature and Sea Level Using Cointegration Methods. *J. Climate*, 25, 7822–7833. Finds temperature and sea level share a common stochastic trend, but the 1940s warming episode deviates from the fitted relationship, attributed to internal ocean variability rather than external forcing — a direct challenge to stationarity assumptions underlying extrapolation.
- Grassi, S., Hillebrand, E., & Ventosa-Santaularia, D. (2013). The statistical relation of sea-level and temperature revisited. *Dyn. Atmos. Oceans* (special issue). Re-estimates Rahmstorf’s proportionality coefficient with methods appropriate for two co-trending series; the sensitivity remains at only marginal statistical significance and the 90% CI on 2100 GMSLR widens to 15–150 cm.

Relevance: Moderate. Not competing frameworks for SPECTRUM, but the manuscript’s brief mention of semi-empirical models should cite 1–2 of these directly rather than relying on the AR5 assessment alone as a secondhand summary of the critique.

3.4 Kopp et al. (2016) — a genuinely Bayesian semi-empirical precedent

Citation: Kopp, R. E., Bittermann, K., Horton, B. P., Donnelly, J. P., Gehrels, W. R., Hay, C. C., et al. (2016). Temperature-driven global sea-level variability in the Common Era. *Proc. Natl. Acad.*

Sci., 113(38).

DOI: 10.1073/pnas.1517056113

Approach. A Bayesian errors-in-variables statistical model relating reconstructed GMST to reconstructed GMSL over the Common Era (the past ~ 2000 years), explicitly propagating measurement and proxy uncertainty through the temperature-sea-level relationship rather than fitting it by ordinary least squares.

Relevance: High. The only paper in the Rahmstorf/Vermeer semi-empirical lineage (Section 3.1) that performs genuine Bayesian calibration with propagated uncertainty rather than a point-estimate regression, making it methodologically closer to SPECTRUM’s Bayesian DOLS approach than Rahmstorf (2007) or Vermeer & Rahmstorf (2009) themselves. Worth citing alongside them wherever the manuscript discusses precedent for Bayesian treatment of the GMST-GMSL relationship specifically.

3.5 Rahmstorf, Foster & Cazenave (2012) — statistical evidence for semi-empirical skill

Citation: Rahmstorf, S., Foster, G., & Cazenave, A. (2012). Evaluating the ability of process based models to project sea-level change. *Environ. Res. Lett.*, 7, 044035.

DOI: 10.1088/1748-9326/7/4/044035

Approach. Statistically compares the predictive skill of semi-empirical (GMST-forced) models against process-based (IPCC AR4-class) models by hindcasting observed GMSLR over the 1990–2011 satellite/tide-gauge record, finding the semi-empirical class tracks observations at least as well and the process-based models systematically underestimate the observed rate.

Relevance: Moderate–high. Direct statistical evidence, external to SPECTRUM, for the core model-complexity argument that simpler, observation-calibrated models can out-predict more complex process-based ones over the historical record. Pairs naturally with Held (2005, Section 6.4) and Aschwanden et al. (2021, Section 6.3) as a third, empirical (rather than purely conceptual) anchor for that argument.

4 Deep uncertainty, MISI/MICI, and WAIS tail-risk framing

4.1 Naughten, Holland & De Rydt (2023)

Citation: Naughten, K. A., Holland, P. R., & De Rydt, J. (2023). Unavoidable future increase in West Antarctic ice-shelf melting over the twenty-first century. *Nature Climate Change*, 13, 1222–1228.

DOI: 10.1038/s41558-023-01818-x — already in the local Zotero library.

Approach. A regional ocean model of the Amundsen Sea embayment shows ice-shelf melt accelerating at similar rates across all emissions scenarios tested this century, i.e., that ocean-driven WAIS melt through roughly 2100 is largely already locked in.

Relevance: Very high. This is the strongest existing, independent, dynamical-modeling support for SPECTRUM’s headline claim that the high-risk tail is largely insensitive to emissions scenario. It should be a central citation for that claim.

4.2 Darnell et al. (2025) — a counterpoint to engage directly

Citation: Darnell, C., Rennels, L., Errickson, F., Wong, T. E., & Srikrishnan, V. (2025). The interplay of future emissions and geophysical uncertainties for projections of sea-level rise. *Nature*

DOI: 10.1038/s41558-025-02457-0

Approach. Using the calibrated BRICK model (Wong et al. 2017, Section 1.3), decomposes SLR projection variance into emissions-driven and geophysical (largely ice-sheet) components as a function of forecast horizon. Finds emissions-scenario uncertainty dominates in the near-to-mid term (specifically the 2065–2075 window), with geophysical/tipping-point uncertainty in Antarctica becoming dominant only later.

Relevance: Very high — requires a direct response. This paper’s headline decomposition is close to a rival claim to SPECTRUM’s: it argues emissions choice *does* materially affect the SLR distribution over multi-decadal horizons before geophysical uncertainty takes over, whereas SPECTRUM emphasizes that emissions cuts move the median but leave the tail (which we attribute predominantly to WAIS) largely unchanged. This surfaced independently from two separate literature searches run for this report, which is a reasonable signal that it is load-bearing rather than a minor citation. The manuscript should engage with it explicitly — for example, by clarifying that our claim is about the shape/tail of the distribution at a fixed horizon (end of century) rather than about total-variance decomposition across horizons, which are not necessarily in conflict, but the distinction should be stated rather than left for a referee to raise.

4.3 MISI/MICI physics and structural uncertainty

- Robel, A.A., Seroussi, H., & Roe, G.H. (2019). Marine ice sheet instability amplifies and skews uncertainty in projections of future sea-level rise. *PNAS*, 116(30), 14887–14892 (DOI 10.1073/pnas.1904822116; in Zotero). Stochastic perturbation analysis showing MISI dynamics structurally produce a right-skewed SLR distribution — theoretical grounding for treating WAIS with a fat right tail rather than choosing one ad hoc.
- DeConto, R.M. & Pollard, D. (2016). Contribution of Antarctica to past and future sea-level rise. *Nature*, 531, 591–597 (DOI 10.1038/nature17145; in Zotero). The foundational MICI/hydrofracturing model producing multi-meter 21st-century Antarctic contributions under high emissions.
- Edwards, T.L., et al. (2019). Revisiting Antarctic ice loss due to marine ice-cliff instability. *Nature*, 566, 58–64 (DOI 10.1038/s41586-019-0901-4). A statistical emulator with properly propagated parametric uncertainty concludes the original DeConto & Pollard projections likely overestimated central/high-end SLR.
- Morlighem, M., Goldberg, D., Barnes, J.M., et al. (2024). The West Antarctic Ice Sheet may not be vulnerable to marine ice cliff instability during the 21st century. *Science Advances*, 10, eado7794 (in Zotero). A multi-model intercomparison arguing MICI is unlikely to activate before 2100 under realistic near-term forcing.
- Bassis, J.N., Berg, B., Crawford, A.J., & Benn, D.I. (2021). Transition to marine ice cliff instability controlled by ice thickness gradients and velocity. *Science*, 372, 1342–1344. Refines the physical trigger condition for MICI, showing the transition to runaway calving is more physically constrained than the original parameterization.

Relevance: High, but mixed. Together these papers show that whether and when MICI activates this century is itself an actively contested question in the ice-sheet-dynamics literature. Edwards (2019) and Morlighem (2024) are important tempering counter-evidence that should be

cited alongside DeConto & Pollard when justifying the probability weight assigned to SPECTRUM’s fast/MICI-type WAIS branch, rather than treating MICI risk as settled.

4.4 Deep-uncertainty decision framing and near-term predictability

- Bakker, A. M. R., Wong, T. E., Ruckert, K. L., & Keller, K. (2017). Sea-level projections representing the deeply uncertain contribution of the West Antarctic ice sheet. *Sci. Rep.*, 7, 3880 (DOI 10.1038/s41598-017-04134-5). Explicitly frames WAIS via deep-uncertainty decision analysis rather than a process model — the closest existing methodological precedent for SPECTRUM’s WAIS scenario mixture and terminology.
- Kopp, R. E., DeConto, R. M., Bader, D. A., et al. (2017). Evolving Understanding of Antarctic Ice-Sheet Physics and Ambiguity in Probabilistic Sea-Level Projections. *Earth’s Future*, 5, 1217–1233 (in Zotero). Shows that alternative structural assumptions about Antarctic physics (with vs. without MICI-type processes) move probabilistic SLR distributions far more than emissions-scenario choice does — close precedent for SPECTRUM’s “orthogonal dimension of risk” framing.
- McCormack, F. S., et al. (2026). Emergent decadal predictability in Antarctic contribution to sea level rise. *Nature*, 654, 609–613 (DOI 10.1038/s41586-026-10614-4). Finds the 2025 Antarctic mass-loss rate is strongly predictive of the rate for the following 3–5 decades across models and scenarios, with predictability breaking down only late-century as marine ice-sheet feedbacks emerge — independent dynamical-modeling evidence converging with SPECTRUM’s rate-anchoring strategy and its finding that most forecast uncertainty is late-century and WAIS-driven.

4.5 Sriver et al. (2012) — a Bayesian-calibrated upper bound

Citation: Sriver, R. L., Urban, N. M., Olson, R., & Keller, K. (2012). Toward a physically plausible upper bound of sea-level rise projections. *Climatic Change*, 115(3–4), 893–902.

DOI: 10.1007/s10584-012-0610-6

Approach. Uses a reduced-complexity, Bayesian-calibrated climate/sea-level model (DOECLIM plus semi-empirical components) with expert-elicited priors on poorly constrained Antarctic parameters to derive a physically plausible upper bound on 21st-century GMSLR, rather than resolving Antarctic dynamics with a process model.

Relevance: Moderate–high. An early precedent for treating the ice-sheet tail as a Bayesian-calibrated bounding problem rather than a process-model output, in the same spirit as SPECTRUM’s WAIS scenario mixture (Section 4.4) though without the explicit deep-uncertainty scenario structure; useful as a citation for the general strategy of bounding the tail statistically when the underlying physics is not well constrained.

5 Observational anchoring, nowcasting, and naive extrapolation

5.1 Closest precedents to the naive-extrapolation baseline

Citation: Nerem, R. S., Frederikse, T., & Hamlington, B. D. (2022). Extrapolating Empirical Models of Satellite-Observed Global Mean Sea Level to Estimate Future Sea Level Change. *Earth’s Future*, 10, e2021EF002290 — already in the local Zotero library.

Approach. Directly extrapolates a linear-plus-quadratic fit of satellite altimetry GMSL to 2050 and compares it against IPCC AR6, finding rough consistency but flagging that the extrapolation assumes constant rate and acceleration.

Relevance: Very high. The closest existing precedent to SPECTRUM’s own naive-extrapolation baseline; should be cited and explicitly differentiated (SPECTRUM extends the horizon to 2100 and treats the naive forecast as one rung in an explicit model-complexity hierarchy rather than a standalone estimate).

Also relevant: Nerem et al. (2018, *PNAS*, DOI 10.1073/pnas.1717312115), the original detection/quantification of GMSLR acceleration from the altimetry record ($\sim 0.084 \text{ mm yr}^{-2}$); Hamlington et al. (2024, *Comm. Earth & Environ.*, DOI 10.1038/s43247-024-01761-5), which quantifies the doubling of GMSLR rate over 1993–2023 underlying the manuscript’s own doubling-time discussion; and Dangendorf et al. (2019, *Nat. Clim. Change*), which shows the acceleration signal predates the satellite era, supporting the robustness of extrapolating it.

5.2 Observational constraint as a general method

Citation: Ribes, A., Qasmi, S., & Gillett, N. P. (2021). Making climate projections conditional on historical observations. *Science Advances*, 7, eabc0671.

DOI: 10.1126/sciadv.abc0671

Approach. A general statistical framework, outside the SLR literature, for constraining long-term CMIP-style warming projections using the full historical observational record, narrowing projection uncertainty by roughly a factor of 2–3.

Relevance: High. The clearest methodological ancestor, outside sea level specifically, for SPECTRUM’s rate-space blending step. Useful as a citation establishing that observationally-constrained projection is an established technique in climate science more broadly, not an ad hoc device specific to this study.

5.3 A necessary nuance

Citation: Törnqvist, T. E., Conrad, C. P., Dangendorf, S., & Hamlington, B. D. (2025). Evaluating IPCC Projections of Global Sea-Level Change From the Pre-Satellite Era. *Earth’s Future*, 13, e2025EF006533.

Approach. Finds that the IPCC Second Assessment Report (1995/96) mid-range GMSL projection verified to within $\sim 1 \text{ cm}$ accuracy over the following 30 years.

Relevance: High — a needed qualifier. SPECTRUM’s critique of process-model divergence from observations is specifically a critique of *AR6*, not of IPCC assessments in general; this paper shows an earlier IPCC vintage performed very well. The manuscript’s introduction should make this scope explicit rather than let the broader claim stand unqualified (cross-referenced in Section 1.5).

5.4 Simplicity-in-forecasting literature

Beyond Green & Armstrong (2015, already cited as Green2015), Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020), The M4 Competition: 100,000 time series and 61 forecasting methods, *Int. J. Forecasting*, 36, 54–74 (DOI 10.1016/j.ijforecast.2019.04.014), is a large-scale empirical forecasting competition confirming that simple statistical and hybrid methods remain highly competitive against complex ones — a useful second, independent citation for the general model-parsimony argument, alongside Held (2005, Section 6.4).

5.5 Pfeffer, Harper & O’Neel (2008) — kinematic constraints

Citation: Pfeffer, W. T., Harper, J. T., & O’Neel, S. (2008). Kinematic constraints on glacier contributions to 21st-century sea-level rise. *Science*, 321(5894), 1340–1343.

DOI: 10.1126/science.1159099

Approach. Bounds plausible 21st-century glacier and ice sheet contributions using observed and physically limiting ice-discharge velocities and geometric constraints (kinematics), rather than resolving ice dynamics with a full process model or fitting a semi-empirical GMST relationship.

Relevance: High. An early, influential precedent for constraining ice-sheet/glacier contributions directly from observed rate and geometric limits rather than modeled physics — the same underlying philosophy as SPECTRUM’s rate- and observation-anchored kinematic extrapolation baseline (Section 5.1), applied a decade and a half earlier with much sparser observational records. Worth citing alongside Nerem, Frederikse & Hamlington (2022) as an early precedent for the kinematic/observational-constraint strategy specifically for ice components.

6 Reduced-complexity discharge and calving parameterizations

6.1 Closest precedent to the Greenland delayed-response model

Citation: Levermann, A., Winkelmann, R., Nowicki, S., et al. (2014). Projecting Antarctic ice discharge using response functions from SeaRISE ice-sheet models. *Earth Syst. Dynam.*, 5(2), 271–293.

DOI: 10.5194/esd-5-271-2014

Approach. Derives linear, time-lagged response (convolution) functions relating ice discharge to ocean-driven basal-melt forcing, calibrated against full ice-sheet-model experiments rather than resolved at run time; this discharge parameterization underlies LARMIP and subsequent ISMIP6-adjacent emulators.

Relevance: High. The closest published precedent to SPECTRUM’s own linear, time-lagged discharge model, $D = \gamma T_{\text{ocean}}(t - \delta) + r_0$ (Eq. ?? in the main text), though Levermann et al. calibrate against model output rather than direct observations as SPECTRUM does.

6.2 Plume-theory melt-rate scaling underlying the discharge sensitivity

Citation: Slater, D. A., Goldberg, D. N., Nienow, P. W., & Cowton, T. R. (2016). Scalings for Submarine Melting at Tidewater Glaciers from Buoyant Plume Theory. *J. Phys. Oceanogr.*, 46(6), 1839–1855.

Approach. Derives the canonical plume-theory melt-rate scaling relating subglacial discharge and ocean thermal forcing to submarine melt rate.

Relevance: Moderate–high. The theoretical basis for the “Slater/Wood-type” melt sensitivities already cited in the manuscript’s dimensional-scaling argument (Eqs. ??–??); several downstream applications of this scaling (Xu et al. 2013, Rignot et al. 2016, Slater et al. 2019) are already cited in the manuscript and in Zotero.

6.3 A candidate primary citation for the model-complexity argument

Citation: Aschwanden, A., Bartholomaus, T. C., Brinkerhoff, D. J., & Truffer, M. (2021). Brief communication: A roadmap towards credible projections of ice sheet contribution to sea level. *The Cryosphere*, 15, 5705–5715.

DOI: 10.5194/tc-15-5705-2021 — already cited in the manuscript as Aschwanden2021, but

currently only for the specific claim that process models diverge from hindcasts; it is also a more general statement of the manuscript’s model-complexity argument.

Approach. Argues that current ice-sheet models are systematically under-constrained relative to their degrees of freedom, and that projection credibility requires matching model complexity to the calibration data actually available.

Relevance: High. Beyond its existing citation, this paper is close to a direct statement of SPECTRUM’s central methodological argument and could be cited a second time, alongside Green (2015) and Held (2005), specifically where the manuscript makes the general complexity-vs.-data-constraint case.

6.4 General model-hierarchy literature

Citation: Held, I. M. (2005). The Gap Between Simulation and Understanding in Climate Modeling. *Bull. Amer. Meteor. Soc.*, 86(11), 1609–1614.

Approach. The canonical argument for a “hierarchy of models” bridging idealized, well-understood low-order models and comprehensive simulation models, arguing that scientific understanding (and often forecast robustness) is better served by simpler models than by ever-more-complex ones.

Relevance: High. The closest general antecedent, outside the SLR or ice-sheet literature specifically, to the manuscript’s own “hierarchy of model complexity” framing (naive → semi-empirical → component-level). Pairs naturally with Green (2015) as a second anchor citation for that argument; see also Jeevanjee, N., Hassanzadeh, P., Hill, S., & Sheshadri, A. (2017), A perspective on climate model hierarchies, *JAMES*, 9(4), 1760–1771 (DOI 10.1002/2017MS001038), a more formally developed follow-on.

7 Items published in approximately the last twelve months (Sept. 2025–Sept. 2026)

Most items in this window are cross-referenced above by topic (Felixson et al. 2025, Darnell et al. 2025, Naughten-adjacent items, Törnqvist et al. 2025, McCormack et al. 2026). Additional recent items not detailed elsewhere:

- Perrette, M. & Mengel, M. (2025). Relative sea level projections constrained by historical trends at tide gauge sites. *Science Advances* (DOI 10.1126/sciadv.ado4506). Generates local SLR projections directly from tide-gauge/GPS/altimetry historical trends rather than from component decomposition; global medians/ranges broadly consistent with AR6. **Relevance: High** as a genuinely different observation-driven alternative to component summation, worth a direct contrast in the framework discussion.
- Coulon, V., Klose, A. K., Edwards, T., Turner, F., Pattyn, F., & Winkelmann, R. (2025). From short-term uncertainties to long-term certainties in the future evolution of the Antarctic Ice Sheet. *Nature Communications*, 16, 10385 (DOI 10.1038/s41467-025-66178-w). Argues short-term (decadal) and long-term (centennial) Antarctic ice-sheet uncertainty are largely decoupled — converges with McCormack et al. (2026, Section 4.4) and supports SPECTRUM’s near-term rate-anchoring logic.
- Winkelmann, R., Garbe, J., Donges, J. F., Albrecht, T., et al. (2026). Mapping tipping risks from Antarctic ice basins under global warming. *Nature Climate Change* (DOI 10.1038/s41558-025-02554-0).

Basin-by-basin analysis of gradual vs. threshold/tipping-point ice loss across Antarctica under different warming levels; complements SPECTRUM’s WAIS scenario mixture with a basin-resolved tipping-point framing.

- Nauels, A., Nicholls, Z., Möller, T., Hermans, T.H.J., Mengel, M., Kloenne, U., Smith, C., Slangen, A.B.A., & Palmer, M.D. (2025). Multi-century global and regional sea-level rise commitments from cumulative greenhouse gas emissions in the coming decades. *Nature Climate Change*, 15(11) (DOI 10.1038/s41558-025-02452-5) — already in Zotero. A direct descendant of Nauels et al. (2017, Section 1.6); quantifies how near-term emissions choices lock in 0.3–0.8 m of GMSLR commitment by 2300, a useful comparator for SPECTRUM’s own emissions-vs-tail-risk framing.

No new IPCC-adjacent government or task-force assessment (NOAA Interagency Task Force, UK Climate Change Committee, Dutch Delta Programme) was identified in this twelve-month window.

8 Summary

Study	Calibration source	Components	Relevance
<i>Core component-summation frameworks (Sec. 1)</i>			
Mengel et al. (2016)	Observations (indep.)	4	Very high
Kopp et al. (2014)	IPCC assessment	5	High
Wong et al. (2017)	Observations (joint)	5	High
Kopp et al. (2023)	Process models	5+	High
Fox-Kemper et al. (2021)	CMIP6 + ISMIP6	5	High
Nauels et al. (2017)	CMIP5 emulation	7	Moderate–high
<i>Additional frameworks (Sec. 2)</i>			
Le Bars & Drijfhout (2025)	Observations (indep.)	6	High
Felikson et al. (2025)	— (perspective)	n/a	Very high
<i>Semi-empirical / non-component (Sec. 3, 5)</i>			
Rahmstorf (2007)	Observations (GMST)	1	Very high
Vermeer & Rahmstorf (2009)	Observations (GMST)	1	Very high
Kopp et al. (2016)	Observations (GMST, Bayesian)	1	High
Grinsted & Christensen (2021)	Observations (GMST)	1	High
Nerem, Frederikse & Hamlington (2022)	Observations (altimetry)	1	Very high
Pfeffer, Harper & O’Neel (2008)	Observations (kinematic)	2	High
Perrette & Mengel (2025)	Observations (tide gauge)	0 (spatial)	High

The core conclusion of the 2026-05-10 search stands: no published framework combines (i) direct Bayesian calibration of each component to observational data, (ii) a structured deep-uncertainty treatment for ice sheet instabilities, and (iii) rate-space blending with satellite-era observations to anchor near-term projections. The 2026-09-03 update strengthens rather than overturns that conclusion, but identifies two papers the manuscript should engage with directly rather than simply add to the bibliography: Darnell et al. (2025), whose emissions-vs-geophysical variance decomposition is close to a rival claim (Section 4.2), and Felikson et al. (2025), whose community-consensus perspective is close to a direct statement of SPECTRUM’s motivating philosophy (Section 2.2).

9 Action items for the manuscript

1. Cite Vermeer & Rahmstorf (2009) — already in `references.bib` as `Vermeer2009` but unused — wherever the semi-empirical class is discussed, and consider citing Grinsted & Christensen (2021) alongside it if the manuscript takes an explicit position on the rate-vs-level question.
2. Add Held (2005) (and optionally Jeevanjee et al. 2017) as a second anchor citation, alongside Green (2015), for the model-hierarchy/complexity argument in the introduction.
3. Add Felikson et al. (2025) to the introduction or discussion as the field-level motivation for observation-constrained projection.
4. Add an explicit sentence engaging with Darnell et al. (2025) where the manuscript makes the “emissions cuts move the median, not the tail” claim, distinguishing SPECTRUM’s fixed-horizon tail-shape claim from their across-horizon variance decomposition.
5. Qualify the “process models diverge from observations” claim (Section 1.5/1) to specify AR6 rather than IPCC assessments generally, citing Törnqvist et al. (2025) as the reason for the qualification.
6. Consider citing Naughten et al. (2023) as the primary independent, dynamical-modeling support for the claim that the WAIS-driven tail is largely insensitive to emissions scenario, alongside Kopp et al. (2017) and McCormack et al. (2026).
7. Consider a brief citation to Levermann et al. (2014) when introducing the Greenland delayed-response discharge model, as the closest published precedent for a linear, time-lagged discharge parameterization.

10 Note on Felikson et al. (2025): why “near-perfect conceptual match”

Felikson et al. (2025) is not a competing forecasting framework in the sense of Sections 1–2 — it makes no projection of its own. What makes it a near-perfect conceptual match is that it is a synthesis, by a broad and senior cross-institutional author group spanning the leading process-based and observational SLR communities, of exactly the argument SPECTRUM is built to demonstrate: that large-uncertainty, physics-heavy component projections should be narrowed using observations, that this requires systematically quantifying uncertainty *structure* (not just magnitude) in both the observations and the models, and that doing so is urgent ahead of the next IPCC cycle. SPECTRUM can be read as one concrete, worked instantiation of the research program that paper calls for at the level of an entire field. That is why it belongs in the introduction or discussion as motivation/context, rather than in the same comparative table as Mengel, Kopp, BRICK, FACTS, or AR6.